

- 8 Fig. 8.1 shows a circuit that produces rectification of an alternating input voltage.

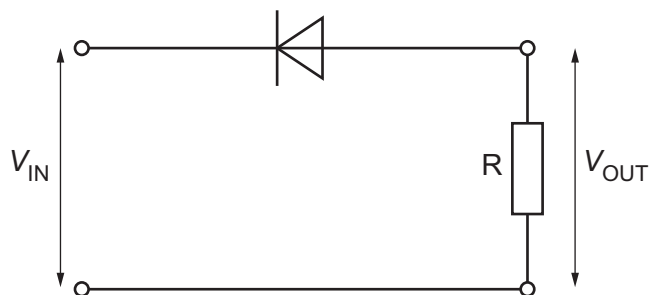


Fig. 8.1

The input voltage V_{IN} is sinusoidal. The rectified output voltage V_{OUT} is applied across resistor R .

The variation of V_{IN} with time t has amplitude V_0 and period T , as shown in Fig. 8.2.

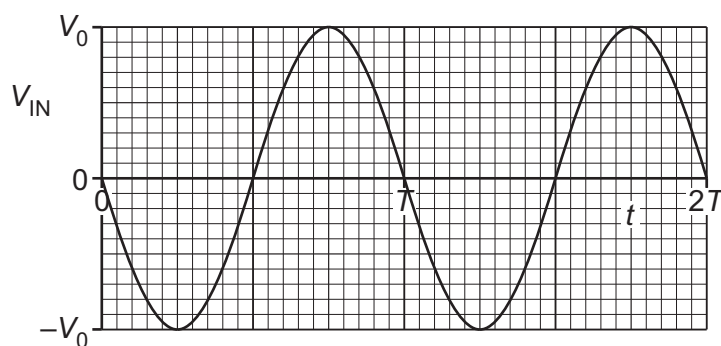


Fig. 8.2

The root-mean-square (r.m.s.) value of V_{IN} is 6.0 V .

- (a) (i) State the type of rectification produced by the circuit of Fig. 8.1.

..... [1]

- (ii) Calculate V_0 .

$V_0 = \dots\dots\dots \text{ V [1]}$

